

IJC437 Week 10: End-to-end Data Science

In this session, you will apply the skills you have developed over the last nine weeks to a real-world scenario that simulates the kind of work you might do as a professional data scientist. Using a recent survey dataset from the UK Data Service, you will clean and transform the data to prepare it for analysis, carry out exploratory analysis, develop predictive models to answer client questions, and produce clear visualisations to communicate your findings to the client.

Note: We are not giving you example code in this worksheet, as you won't generally have examples to work from in practice. Make use of the examples in previous worksheets and your own solutions as you solve the exercises!

Exercises for you to complete are marked with 

Learning objectives. At the end of today's session, you should be able to:

1. Combine the technical skills you have learned for data wrangling, data analysis, data mining, and data visualisation to approach a new data science problem
2. Use dataset documentation and codebooks to understand the features of a new dataset

1. Scenario

You are a professional data scientist working for a consulting firm here in Sheffield. You have just been contracted by HM Treasury to analyse the results of the 2019 edition of the British Social Attitudes Survey, conducted by the University of Manchester.

Hint: You can learn more about this dataset via the UK Data Service at this link:

<https://beta.ukdataservice.ac.uk/datacatalogue/studies/study?id=8850#!/details>

The 2019 version of the survey focused on measuring British people's attitudes towards poverty and welfare. HM Treasury would like to better understand public opinion to guide the policy decisions they need to make around public sector funding.

2. Setting up the dataset

We have posted three files to Blackboard for this dataset, all located in the same Practical folder as this worksheet:

1. **Poverty and Welfare 2019 - Dataset** :: The tab-separated data file.
2. **Poverty and Welfare 2019 - User Guide** :: This PDF describes the collection of the dataset, the principles behind how it was collected, and some high-level summaries of its contents.
3. **Poverty and Welfare 2019 - Codebook** :: This PDF describes each field (column) in the dataset, including what type of variable it is, the distribution of values, and indicators for missing values.

Note: These are the same documents you can access via the UK Data Service link above, however doing so requires you to create an account. We've posted the files without modification to Blackboard instead.

The first step for any working data scientist is to **learn about your data**:



Exercise 1: Download the dataset file, the User Guide, and the Codebook. Load the dataset into R and explore its values using the different tools you've learned about (the Viewer, the summary() method, etc.). Read the description of the variables on page 6 of the User Guide and read about the variable values in the Codebook.

Past work reference: Take a look at the Week 4 lab session (particularly Section 4) for some relevant methods.

Hint: In the same Practical folder as the dataset and this worksheet, you'll also find a link to a [Google Sheet](#) we've created, with columns for you to make notes on the different variables in the dataset. You can use the "Make a Copy" function in Google Sheets to make your own editable version of the worksheet that you can fill in. You're not required to do this, but you may find it helpful for the rest of the worksheet!

Now that you've learned about the data you'll be working with, you can plan your strategy for analysis.

3. Preparing the data for analysis

As you may have noticed in the codebook, most variables in the dataset have some missing data. Missing data are very common when working with real-world datasets: people may not respond to every question, there may be data entry errors, not everyone gets the same surveys, etc.

There are two main questions you need to deal with in practice:

- For which variables is missing data acceptable, and for which variables is it not acceptable?
- What strategy will you use to deal with the missing data?

The easiest strategy is to drop every row that has missing data. Let's try that:



Exercise 2: Using the codebook, identify the set of all values that are used to indicate missing data. Try removing all rows that have any missing values (*Hint: try using na.omit() on the overall dataframe, do not try to do this on a per-column basis*).

- How many rows are left after removing all the missing values? (*Hint: you should get a pretty small number!*)
- Do you think that dropping the missing values is the right call for this dataset?

Past work reference: See the *IJC437 Week 4 lab* session for approaches to handling missing values.

There are many other strategies for dealing with missing data, such as Maximum likelihood, imputation, and more. Selecting a suitable strategy depends on the type of data you have, the reasons it may be missing, and what you want to do with it.

In this practical, we will deal with missing values on a *per-column* basis, depending on what we want to do in each exercise.

Further learning: To learn more about handling missing data and imputation, see these tutorials:

<https://jamescheshire.github.io/learningR/missing-data.html>

<http://juliejosse.com/wp-content/uploads/2018/06/DataAnalysisMissingR.html>

In this case, you've spoken to HM Treasury and one of the primary things they want to know is how different results break down by respondents' age. So, we'll focus on cleaning up the age column:



Exercise 3: Using the User Guide and Codebook, identify the column of the dataset that reports respondents' age.

- Identify the special values used for Missing Values in this column, and remove rows with those values from the dataset.
- How many rows did this affect?
- Plot a histogram of the age values. Does this seem like an even distribution?

Past work reference: Take a look at the *Week 4 lab* session for dealing with missing data, and *Week 3 lab* session for histograms.

The *resolution* (i.e. the specificity of the values) of the age column is more than HM Treasury is interested in. They would like to see the results broken down by only 4 age groups:



Exercise 4: Use the Codebook to identify how the age variable is scaled. Re-scale this variable from 1-4 instead, such that:

1 = 18-34

2 = 35-54

3 = 55-64

4 = 65+

Plot a histogram with these new age values. Does the distribution seem more even than before?

Past work reference: Take a look at the Week 4 lab session (particularly the `mutate()` function) as one strategy for this, and Week 3 for histograms.

4. Exploratory data analysis

Now that the data are prepared, it's time to get familiar with what you're working with and start answering client questions.

Based on demographic trends in the country, HM Treasury hypothesises that there is a correlation between age and education level in this dataset. You want to put this to the test.

First, let's just get a sense of what the distribution looks like:



Exercise 5: Use the dataset documentation to identify the column for education level.

- Copy the dataset and drop rows with missing values in this column.
- Now count up the number of rows with each potential value for education level, and break it down by age (i.e. a matrix where age is the rows, education is the columns). Remember to use the re-scaled values for age you set above!
- Based on what you're seeing in this distribution, do you think there is a correlation between age and education level?

Past work reference: Take a look at the Week 4 lab session for missing data. For totaling counts for each value, look online for “frequency table” calculation. For correlation testing, see Week 5 lab session.

Let’s ask that last question more formally, so you have something firm to tell the client. First we’ll reformulate their intuition as a testable hypothesis:

H_0 : there is no correlation between Age and Education level

H_a : there is a correlation between Age and Education level

Now let’s test it, using chi-squared because these are ordinal (discrete) variables:



Exercise 6: Using the dataset copy you created in Exercise 5 (where you dropped missing values for education level):

- Run chi-squared (*Hint: use `chisq.test` function*) to test the correlation between age and education level.
- Does the p-value you get suggest you should reject the null hypothesis, or fail to reject it?
- Use a box plot to visualise the distribution of education level (y-axis) by age (x-axis).

References: Look online for guidance on [chisq.test](#) and for using `ggplot2` to generate box plots.

5. Developing predictive models

Your client (HM Treasury) has a hunch that attitudes towards fraud in government benefits schemes are linked to political preferences, but that this relationship is mediated by income level. You know that the `NatFrEst` variable in your dataset helps to measure these attitudes, since it reflects people’s estimate of how common fraud is among beneficiaries. So let’s use the data you have to put this idea to the test:



Exercise 7: Train a linear regression model to predict NatFrEst based on politics and income. You'll need to do the following steps:

1. Create a copy of the dataset and remove rows with missing values for NatFrEst (use the codebook to find what these are)
2. Income level is represented by the eq_inc_quintiles variable, and this also has some missing values. Drop the rows with missing values for this variable as well.
3. Train a linear regression model to predict NatFrEst using these variables: libauth, lefttrigh, welfare2 (all three of these describe political leanings), Politics (the person's engagement in politics), and eq_inc_quintiles.

What is the R^2 of the model? What if you only use the political leaning variables: libauth, lefttrigh, welfare2? What do these results say about the original hypothesis?

Past work reference: Take a look at the Week 5 lab session for training and evaluating linear regression models.

The client also thinks that attitudes towards government tax and spending are likely to vary based on respondents' age, political leanings, and engagement with politics. Let's see how that pans out:



Exercise 8: Fit a logistic regression model to classify whether a respondent thinks that spending should be increased or not, based on the TaxSpend variable. You'll need to go through the following steps:

1. Drop the rows with missing values and the "None" category for the TaxSpend variable (Hint: use the codebook!)
2. Combine the "Reduce taxes and spend" and "Keep taxes and spend" responses into a single class (it's really "Don't increase taxes and spend", but just label it 0), so you're making a binary prediction between "Increase taxes and spend" (1) and "Don't increase taxes and spend" (0).
3. Randomly shuffle the rows in the dataset and split into 70% for training and 30% for testing.
4. Fit a logistic regression model to make this prediction, using the variables, libauth, lefttrigh, welfare2, Politics, and age (using the rescaled variable you created earlier). Only use the training data!
5. Use the trained model to make predictions on the test set and calculate accuracy.

How accurate is the model? Does this support the client's hypothesis about people's TaxSpend opinions?

Past work reference: Take a look at the Week 5 lab session for training and evaluating logistic regression models.

6. Communicating context

As you've been talking with the client, you've noticed they've talked a lot about how people use social media, and about political differences in social attitudes.

You're now preparing your report to share with them, and you know that giving them some extra descriptive analyses will help you win the next contract too. So let's look at how people engage with social media, first:



Exercise 9: The SMNews variable describes how often people read the news via social media. Create a heatmap to illustrate the relationship between SMNews and PartyId2 (make sure you drop missing values for both first!). Do you see any interesting patterns?

Create a second heatmap looking at SMNews and Spend1 (priorities for spending). Make sure to drop missing values! What patterns do you see here?

Past work reference: See [here](#) and [here](#) for methods to create and interpret heatmaps.

To wrap up, let's get some nice takeaway visualisations of overall correlations between social attitude variables and political party affiliation:



Exercise 10: Visualise the correlation matrix for the following variables: TaxSpend, SocialTrust, MuchPov, Poverty1, Poverty2, and PartyId2. Do you notice any interesting patterns?

Past work reference: Take a look at the Week 7 lab session for more on visualising and interpreting correlation matrices.



Exercise 11: A large part of a data scientist's work is summarising your findings to share them with others.

Write 1-2 paragraphs for yourself reflecting on:

- What did you learn about the British Social Attitudes dataset in this exercise?
- What did you learn about applying your data science skills in practice?

Please submit your thoughts along with your feedback on the exercise in the [Post-Practical Google Form survey](#).

WELL DONE! You've just lived a day in the life of a working data scientist.

Further Resources:

- [Missing data](#) in R
- [Linear Regression](#) in R
- [Logistic Regression](#) in R